

Third International Conference on Computing and Network Communications (CoCoNet'19)

OFDM Based RADAR-Communication System Development

Samarendra Nath Sur^{a*}, Prashant Sharma^a, Hardick Saikia^a, Subhankar Banerjee^a,
Arun Kumar Singh^a

^a*Department of Electronics and Communication Engineering, Sikkim manipal Institute Of Technology, Sikkim Manipal University, Majitar, Rangpo, East Sikkim-737136*

Abstract

This paper highlights the hardware implementation of orthogonal frequency division multiplexing (OFDM) based joint radar and communication (RADCOM) system for the vehicular application. Improvement of the communication system due to the application of OFDM is well established and the same advantages can be exploited to enhance the radar performance. The frequency diversity of the OFDM signal improves the radar performance significantly. This motivates the authors to develop a RADCOM system based on OFDM waveform. This paper includes the performance analysis of the OFDM based RADCOM system under the dynamic vehicular channel conditions. Communication system performance is analyzed based on the bit error rate (BER) and the performance of the radar system is evaluated based on target detection.

© 2020 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the Third International Conference on Computing and Network Communications (CoCoNet'19).

Keywords: OFDM, BER, RADAR, Correlation.

1. Main text

The demand for the bandwidth is rising day by day, which is required by the increasing requirement of consumers for high quality communication and sensing ability. We are using a single hardware for communication and radar application, which can reduce the problem of limited spectral resources.

This system is based on combining radar and communication in single system, as a joint system it will add a

* Corresponding author. Tel.: +91-9433745766

E-mail address: samar.sur@gmail.com/samarendra.s@smit.smu.edu.in

unique feature for future intelligent transportation system. With the development of OFDM waveform for communication system, open up an opportunity of new generation digital radar. A significant amount of research work has been going on for the development of the OFDM radar, which has a huge scope in future applications such as target detectability and tracking, [1,2] direction of arrival (DOA) estimation [3] and synthetic aperture radar (SAR) imaging [4-7] fields, etc. As presented in [8-9], OFDM is proven to be suitable for the radar application [8] and the achievability of integrating communication function in radar network. Waveform based on OFDM are well established in communication system [10]. OFDM has the unique advantage of having frequency diversity. And this diverse nature helps the system to encounter the severe wireless channel phenomenon, such as multipath effect, interference effect etc. And the work is going on to exploit the advantages of OFDM for radar. For the radar application, a processing scheme to retrieve two-dimension range velocity image was introduced in [11-13]. The exploitation of multiple frequencies in sensing applications, such as remote sensing of clouds [14], detection of landmines [15], target-classification [16], interpretation of an urban scene [17], etc are well investigated in different forum. The main reason for using OFDM is to exploit its frequency diversity. And the same is highly beneficial for radar as the target behave differently for different frequency, and that enable us to observe the response of the target on the ambiguity function. This paper includes the analysis of the system based on the BER performance and the target range imaging.

Rest of the paper is organized as follows, section 2 represents a simulation model for the joint radar communication system along with system descriptions and simulated results. Section 3 represents the experimented results and corresponding discussions. At last, section 4 concludes the paper.

2. MATLAB Simulation

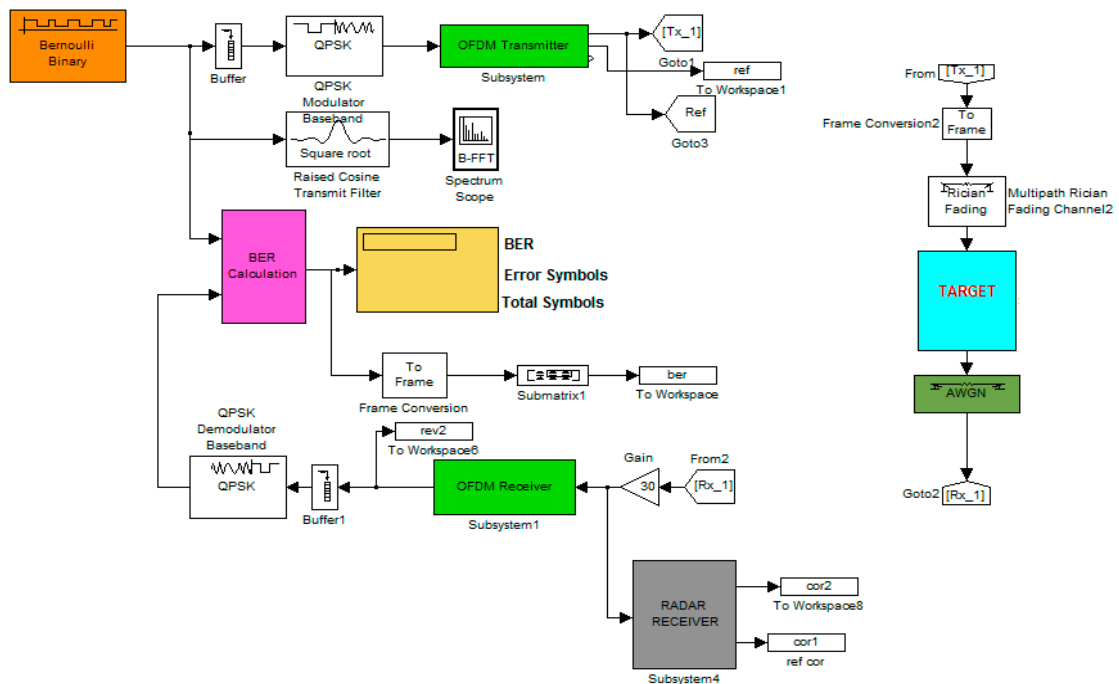


Fig. 1. Simulation Model for OFDM based RADCOM System.

Figure 1 depicts a combine model for join communication and radar system. Based on the signal processing algorithm, one can extract the information related to the vehicle to vehicle (V2V) communication and the ranging

information between the vehicles.

2.1. Communication System.

In the transmitter section, the information bits are modulated using M-ary phase-shift keying (MPSK) modulator and then symbols are then feed to the OFDM modulator. Inverse Fast Fourier transform (IFFT) and Fast Fourier transform (FFT) are utilized to realize the OFDM modulator and demodulator. In order to minimize the ISI formation, cyclic prefix (CP) is concatenated with the OFDM symbol. Then, at the receiver end, channel estimation and equalization have been implemented using zero forcing (ZF) detection. After that the equalized data passed to the demodulator block. As in the figure, in the channel section, authors have considered Rayleigh multipath channel, AWGN channel and radar target model. As radar target model, we have taken the path loss due to the variation in distance and the reflectivity of the target (vehicle). Therefore for the paper includes severe multipath condition both for communication and radar for vehicular application.

2.2. RADAR Signal Processing:

In this model the radar signal processor (RSP) is used to extract the ranging information between two targets. This ranging information is very important in order to minimize the road accident. The range information is carried out based on the correlation processing as discussed below. The receiver for the developed radar is based on the correlation processing. The correlation output represents the existence of the target and the delay/amplitude represent the target range/reflectivity.

The range of the target can be calculated by measuring the delay (T_d) as mentioned in the following formula:

$$R = C \cdot T_d / 2 \quad \text{----- (1)}$$

Where R is the target range, C is the light speed and T_d is the time span for the signal to hit the target and back. A typical correlation processing is explained in the figure 2 as mention below.

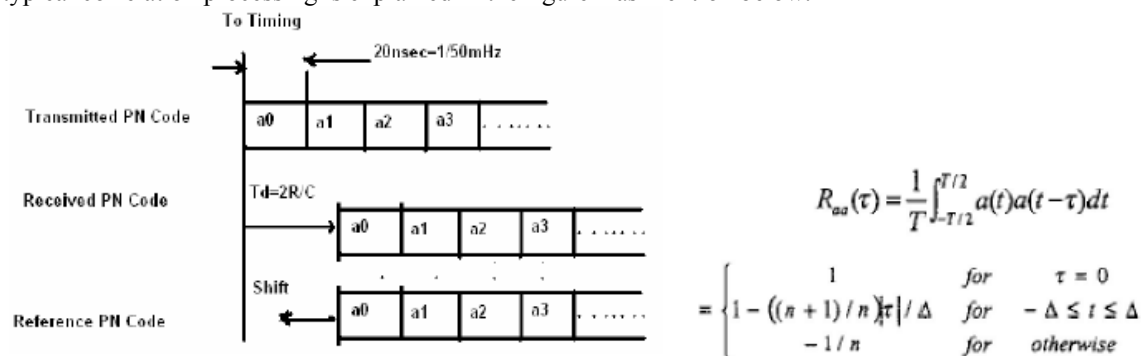


Fig. 2. Correlating PN codes

Here the target detection is completely based on the correlation processing of the PN code, as shown in Fig.3. As in figure 3, the main lobe is actually containing the target information whereas the side lobe mostly perturbed by the clutter and multipath effect. Here, the frequency diversity scheme come into play as it can reduce the multipath effect quite effectively.

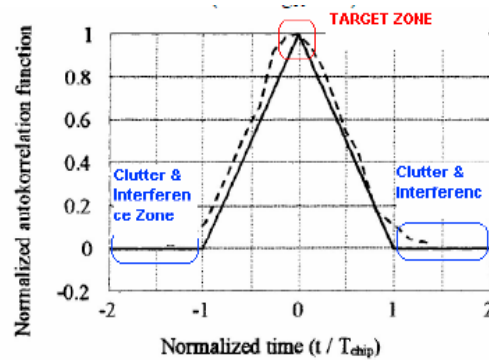


Fig. 3. Calculated and measured auto-correlation function

AS in figure, the correlated peak position on the time axis indirectly provides the target distance information and the amplitude of the correlation peak provide the information related to the target reflectivity. Higher the correlated value corresponds to the bigger target/ vehicle size and lower value for the small size target. And also the correlation processing has the capability to detect multiple target but its resolution depends on the bandwidth of the signal.

2.3. Simulation Results:

In this section, authors have represented the MATLAB simulated result corresponding to the OFDM based communication system as shown in the figure 1.

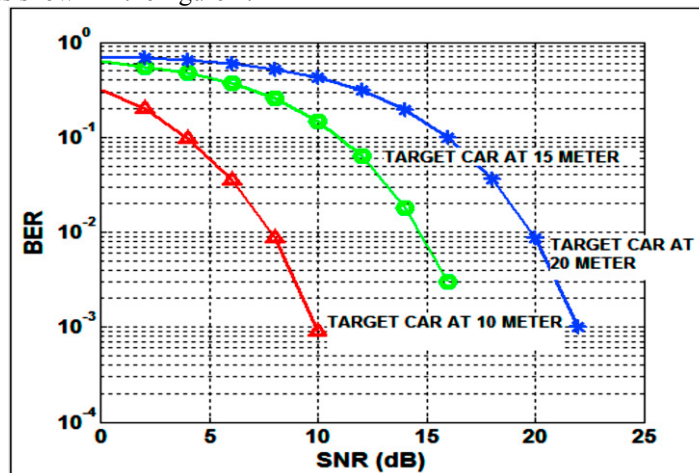


Fig 4. SNR vs BER curves with the change in target car distance.

Above figure 4 shows the BER performance of the RADCOM system based on the OFDM with the change in target vehicle distance. Authors have assumed that the entire system is installed in the user vehicle and the front vehicle is acting as the target vehicle. As in road condition, due to the presence of the multiple vehicle, the channel can be considered as the severe multipath channel. And for the simulation, authors have considered all the necessary conditions. With the change in target (vehicle) distance from the user vehicle, the received signal power degraded which degrade the system BER performance. And that can be taken care by using adaptive power control and that is out of the scope of this paper.

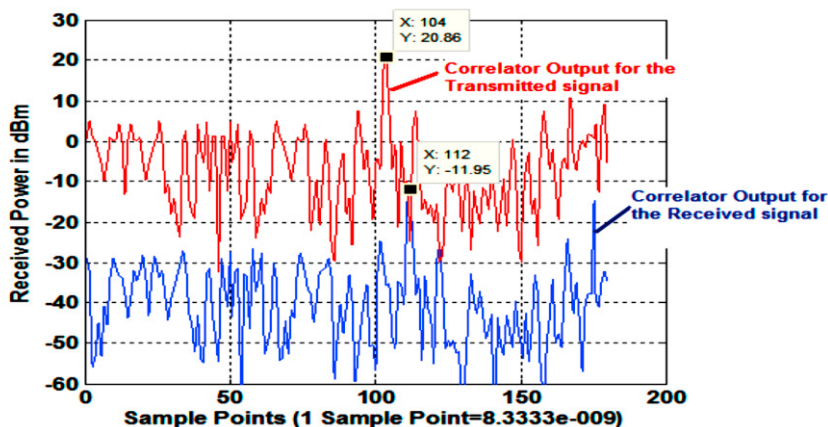


Fig 5. Correlator output of the received signal with target at 10 meters.

The basic approach for the range measurement is to deduct the system delay information from the target returned signal delay information. Here, the system delay information has been calculated using the auto correlation of the reference signal. Figure 5 shows the correlation of the reference and the received signal. From the figure it is evident that the sample difference between the two-correlation peak is 8 samples, which corresponds to an equivalent delay of $8 \times 8.33 \times 10^{-9}$ sec. Therefore, the estimated target range is 9.9960 meter.

3. Hardware Implementation

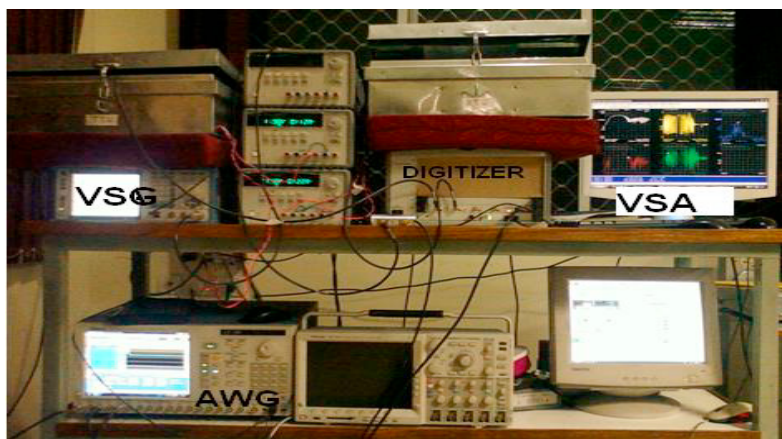


Fig 6 Hardware setup

Above figure 6 shows the transmitter and the receiver section of the OFDM based RADCOM system. Arbitrary Waveform Generator (AWG) is used to have the OFDM modulated signal over the IF of 70 MHz. In AWG, there is an OFDM design panel where user can configure their own system. In this paper, authors have taken the following parameters as shown in figure 7 for the system development. For the up conversion and the down conversion, Vector Signal Generator (VSG) has been used as local oscillator. Here, we have used S-band as the frequency band of operation. Author have used horn antenna for the transmitting and the receiving purpose. In Vector Signal Analyser (VSA), the IF signal is first digitized and then the digitized data feed to the analyser software. Signal triggering, phase synchronization and correlation processes have been implemented using the VSA software.

Define selected symbol

Spacing between Carriers: 200.000 k Hz

☐ User-defined Frequency Weight: C:\Program Files\Tektronix\RFExpress\System\Sam Browse...

Subcarrier table

☒ - to + ☐ all +

Add new subcarriers: 254 Add

Subcarrier	Type	Modulation	Base Data
-128	Data	QPSK	Base Data 1
-127	Data	QPSK	Base Data 1
-126	Data	QPSK	Base Data 1
-125	Data	QPSK	Base Data 1
-124	Data	QPSK	Base Data 1
-123	Data	QPSK	Base Data 1
-122	Data	QPSK	Base Data 1
-121	Data	QPSK	Base Data 1
-120	Data	QPSK	Base Data 1

Edit selected subcarrier(s)

Type: Data Modulation: QPSK Base Data: Base Data 1

-128 Apply Delete

Example: -20, 15, 16-23

Padding/Prefix

☒ Zero Padding ☐ Cyclic prefix

☒ Time: 1.000 u s

☐ Percent: 150.00 %

Frequency Offset: 0 Hz

☐ Clipping Ratio: 1.00

☐ Tone Nulling...

Symbol Information

FFT sampling rate: 51.2 MHz

Subcarrier count: 256

Data carriers: 255

Guard carriers: 0

Pilot carriers: 0

Null carriers: 1

Fig 7 OFDM Parameters.

Table 1 Summary of OFDM Parameter

Number of Subcarrier Spacing	256
Subcarrier spacing	200 KHz
Modulation	QPSK
Bandwidth	51.2 MHz

Figure 7 and Table 1 represent the parameters used for implementing the OFDM system.

The basic system parameter for the baseband section is presented in figure 7 and table 1. Here authors have utilized 256 carries but using this hardware we can go up to 1024 number of carriers.

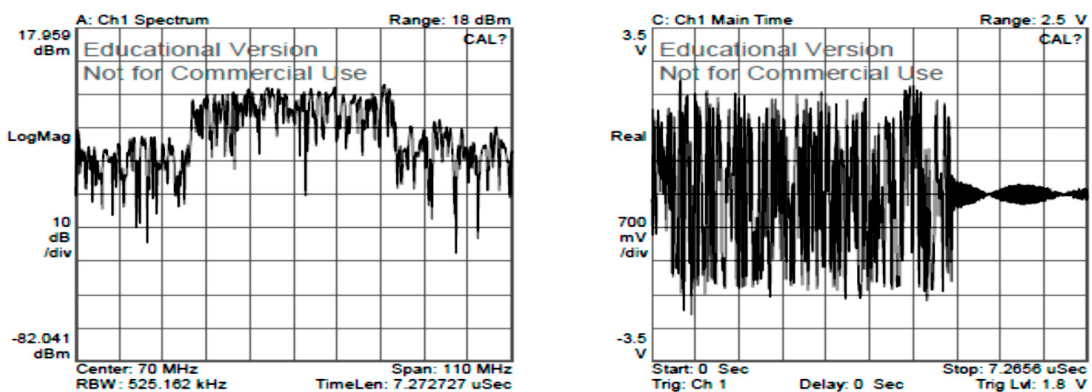


Fig. 8 Transmitted Signal spectrum over IF frequency of 70 MHz and Baseband Time domain signal.

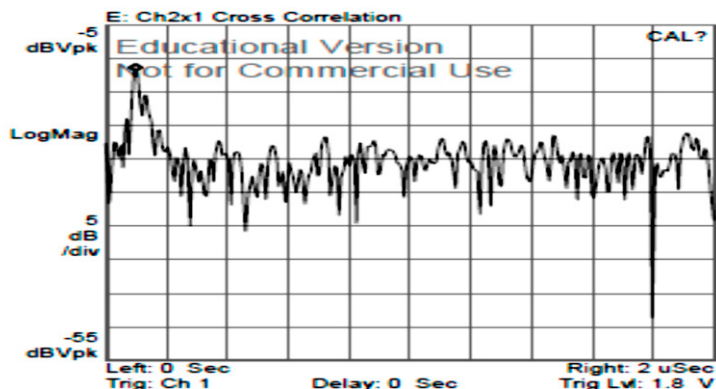


Fig 9 Correlation between transmitted signal and received signal in closed loop (Measurement of system delay).

As a first step, the system delay has been calculated in closed loop manner. The generated signal is passed through the transmitter chain and fed to the receiver chain. Then the received signal is correlated with the reference signal. The total system delay is found out by noting down the position of the correlation peak over the time axis. As in figure 9 the correlation peak is at **99.431 nSec** with **-11.341 dBVpk** magnitude.

After the measurement of the system delay, authors have conducted the actual experimentation for the system validation. For that, target range have been estimated through the cross-correlation processing as presented in the following sections.

5.3.1 Target at 3.5 meter.

Here, target has been placed 3.5 meter in front of the radar system. As a target we have considered a small plat plate.

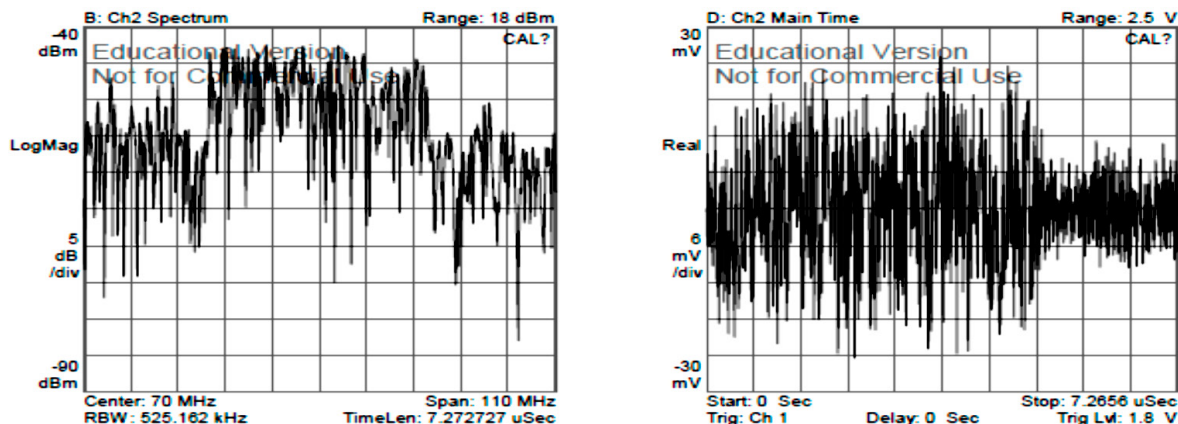


Fig 10 Received Signal spectrum over IF frequency of 70 MHz and Baseband Time domain signal.

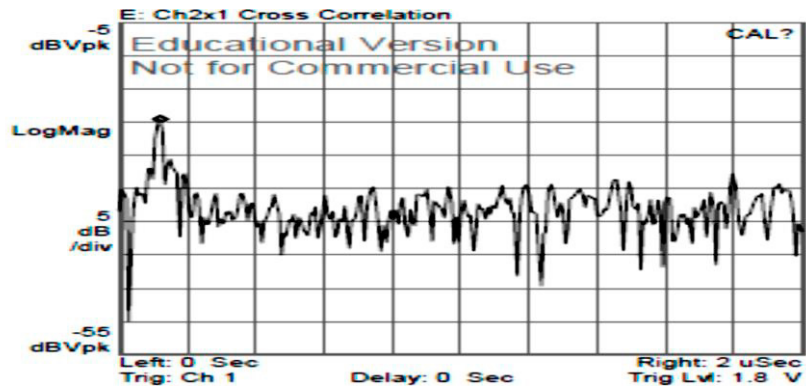


Fig 11 Correlator output

As in figure 11 the correlated output is on 120.738 nSec with the magnitude of -19.548 dBVpk. Therefore the measured target range = $(120.738 - 99.431) \times 10^{-9} \times 3 \times 10^8 \times 0.5 = 3.1961$ meter.

5.3.2 Target at 4.5 meter

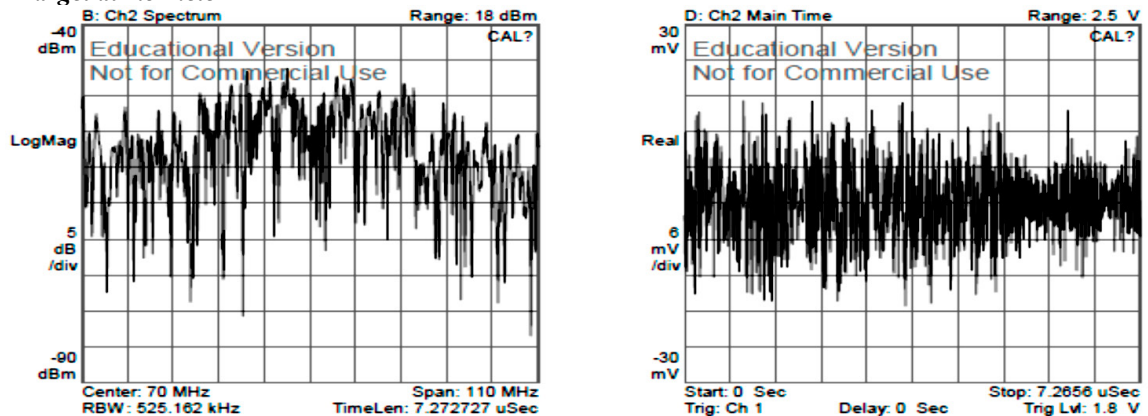


Fig. 12 Received Signal spectrum over IF frequency of 70 MHz and Baseband Time domain signal.

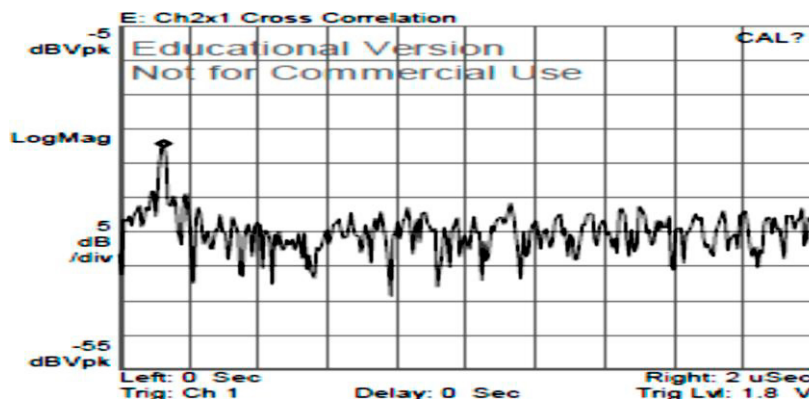


Fig 13. Correlator output

As in figure 13 the correlated output is on 127.840 nSec with the magnitude of -22.159 dBVpk. Therefore the measured target range = $(127.840 - 99.431) \times 10^{-9} \times 3 \times 10^8 \times 0.5 = 4.2614$ meter

4. Conclusion:

This paper represents the performance analysis of a joint radar communication system based on OFDM. Through the experimental results one can conclude that OFDM technique is very much suitable for the radar application also. As the improvement in the communication system performance due OFDM is well established, more emphasis has been given over its target detection performance in this paper. As presented in this paper, the OFDM based radar is able to detect the target and able to estimate the range very close to the actual range. And it proves the utility of the OFDM based joint radar-communication system.

References

- [1]. Sen S, Nehorai A. Adaptive OFDM radar for target detection in multipath scenarios. *IEEE Trans Signal Process* 2011;59(1):78–90.
- [2]. T. Bai, H. Hu and X. Song, "OFDM MIMO radar waveform design with high range resolution and low sidelobe level," 2017 IEEE 17th International Conference on Communication Technology (ICCT), Chengdu, 2017, pp. 1065-1069.
- [3]. Wu XH, Kishk AA, Glisson AW. MIMO-OFDM radar for direction estimation. *IET Radar Sonar Navig* 2010;4(1):28–36.
- [4]. Garmatyuk D, Brenneeman M. Adaptive multicarrier OFDM SAR signal processing. *IEEE Trans Geosci Remote Sens* 2011;49 (10):3780–90.
- [5]. Zhang TT, Xia XG. OFDM synthetic aperture radar imaging with sufficient cyclic prefix. *IEEE Trans Geosci Remote Sens* 2015;53 (1):394–404.
- [6]. Sturm C, Wiesbeck W. Waveform design and signal processing aspects for fusion of wireless communications and radar sensing. *Proc IEEE* 2011;99(7):1236–59. 10. Sturm C, Braun M, Zwick T, Wiesbeck W.
- [7]. G. E. A. Franken, H. Nikookar, P. van Genderen, "Doppler tolerance of OFDM coded Radar Signals, *Proc. 3rd European Radar Conference*, pp. 108 – 111, Manchester, Sept. 2006.
- [8]. C. G. Raghavendra, K. Nagabushan Bhat, M. P. Raghu Srivastava, Rahul N Murthy, P. Vignesh Nayak, N. N. S. S. R. K Prasad, "A novel approach to generate OFDM radar signals", *Electrical Electronics Communication Computer and Optimization Techniques (ICEECCOT) 016 International Conference on*, pp. 141-145, 2016.
- [9]. D. Garmatyuk, J. Schuerger, Y. Morton, K. Binns, M. Durbin, and J. Kimani, "Feasibility study of a multi-carrier dual-use imaging radar and communication system," in *Radar Conference, 2007. EuRAD 2007. European*, Oct. 2007, pp. 194 –197.
- [10]. T. Hwang, C. Yang, G. Wu, S. Li, and G. Ye Li, "OFDM and its wireless applications: A survey," *IEEE Trans. Veh. Technol.*, vol. 58, no. 4, pp. 1673–1694, May 2009.
- [11]. R. F. Tigrek, W. J. A. de Heij and P. van Genderen, "Multi-carrier radar waveform schemes for range and Doppler processing," in *Proc2009 IEEE Radar Conf.*, pp. 1–5.
- [12]. C. Sturm, E. Pancera, T. Zwick, and W. Wiesbeck, "A novel approach to OFDM radar processing," in *Proc. 2009 IEEE Radar Conf.*, pp. 1–4.
- [13]. C. Sturm, M. Braun, T. Zwick, and W. Wiesbeck, "A multiple target Doppler estimation algorithm for OFDM based intelligent radar systems," in *Proc. 7th Eur. Radar Conf.*, 2010, pp. 73–76.
- [14]. Xuanxuan Tian, Zhaohui Song, "On radar and communication integrated system using OFDM signal", *Radar Conference (RadarConf) 2017 IEEE*, pp. 0318-0323, 2017.
- [15]. P. V. Genderen, P. Hakkaart, J. V. Heijenoort, and G. P. Hermans, "A multi frequency radar for detecting landmines: Design aspects and electrical performance," in *Proc. 31st European Microwave Conf.*, vol. 2, London, England, Sep. 24–26, 2001, pp. 249–252.
- [16]. C. Bennett and J. Toomey, "Target classification with multiple frequency illumination," *IEEE Trans. Antennas Propag.*, vol. 29, no. 2, pp. 352–358, Mar. 1981.
- [17]. M. L. Bryan, "Interpretation of an urban scene using multi-channel radar imagery," *Remote Sensing of Environment*, vol. 4, pp. 49–66, 1975.